Global Sales Performance

Executive Report for Stakeholders

Module 4 Capstone Project: Data Cleaning, Analysis and Recommendations

Total Revenue	Total Profit	Units Sold	Profit Margin
$118.73M	**$16.89M**	**1.13M**	**14.2%**

Data: 700 sales transactions across 5 market segments, 5 countries and 6 products, September 2013 to December 2014. Revenue is Sales (Gross Sales less Discounts). All figures are in US dollars.

1. Executive Summary

The business generated **$118.73M in revenue**, **$16.89M in profit** (a 14.2% margin) and **1.13M units**. Sales are spread evenly across the five countries, but profit depends heavily on one segment, and discounting and Enterprise pricing are costing the company money.

Key findings

- **Government is the profit engine.** It brings 44% of revenue but 67% of profit, at a 21.7% margin.
- **Enterprise loses money.** It earns $19.61M in revenue but a loss of $0.61M. It is the only segment with losses (58 transactions), and every country is affected.
- **Discounts erode profit.** High-discount deals make up 31% of revenue but only 20% of profit. Their margin is 9.1%, against 17.9% for low-discount deals.
- **Small Business is large but thin.** It holds 36% of revenue but earns only a 9.8% margin.
- **Channel Partners and Midmarket are very efficient but tiny.** Together they are 3.5% of revenue, with margins of 73% and 28%.
- **Countries are balanced.** Each contributes 18-21% of revenue. France and Germany have the best margins (about 15.5-15.7%) and the United States the lowest (12.0%).
- **Q4 is the peak.** October and December 2014 each reached about $12M. Sep-Dec 2014 revenue was 37% higher than Sep-Dec 2013, and profit was 40% higher.

Priority recommendations

- Fix Enterprise pricing and set a minimum margin for every deal.
- Put an approval limit on high discounts, starting with Enterprise and Small Business.
- Protect Government contracts, and grow Channel Partners and Midmarket.

2. Data Cleaning and Preparation

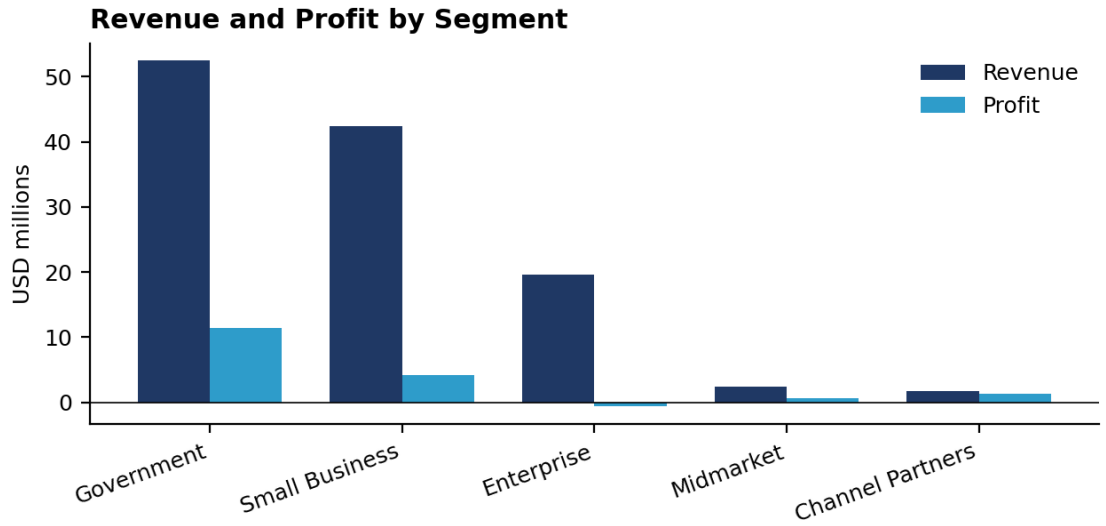
The dataset was cleaned and checked before any analysis. The full Python code is provided in *sales_analysis.py*.

Check	Finding	Action
Size	700 rows, 16 columns	No rows removed
Column header	" Sales" had a leading space	Trimmed to "Sales"
Missing values	53 blank Discount Band values; all have Discounts = 0	Not true missing data. Filled with "None"
Other columns	0 missing values	No action
Duplicates	0 duplicate rows	No action
Data types	Dates, integers and money fields correct	Formatted as date, number, currency
Units Sold	36 rows contain half units	Kept and flagged for source check
Logic checks	Gross - Discounts = Sales; Sales - COGS = Profit; Units x Price = Gross	0 failures in 700 rows
Negative profit	58 rows, all in Enterprise	Kept as real losses
Date coverage	Sep 2013 to Dec 2014 (2013 has 4 months only)	Use like-for-like comparisons

3. Performance Analysis

3.1 Market segments

Segment	Revenue	Profit	Margin	% of Revenue	% of Profit
Government	$52.50M	$11.39M	21.7%	44.2%	67.4%
Small Business	$42.43M	$4.14M	9.8%	35.7%	24.5%
Enterprise	$19.61M	-$0.61M	-3.1%	16.5%	-3.6%
Midmarket	$2.38M	$0.66M	27.7%	2.0%	3.9%
Channel Partners	$1.80M	$1.32M	73.1%	1.5%	7.8%
Total	**$118.73M**	**$16.89M**	**14.2%**	**100.0%**	**100.0%**



Government delivers two thirds of all profit. Enterprise is the only segment below zero. Channel Partners earn a 73% margin but on only $1.8M of revenue.

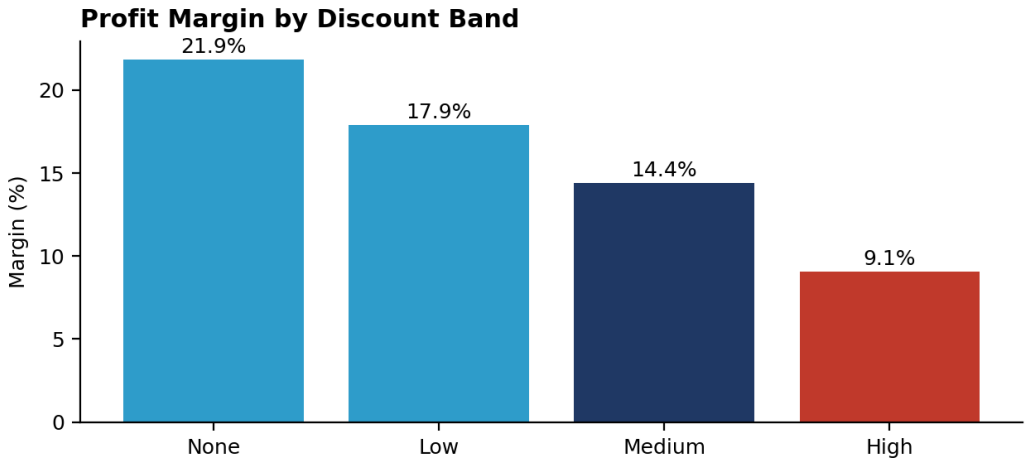
3.2 Countries

Country	Revenue	Profit	Margin	% of Revenue
United States	$25.03M	$3.00M	12.0%	21.1%
Canada	$24.89M	$3.53M	14.2%	21.0%
France	$24.35M	$3.78M	15.5%	20.5%
Germany	$23.51M	$3.68M	15.7%	19.8%
Mexico	$20.95M	$2.91M	13.9%	17.6%

Revenue is balanced across markets. The United States has the lowest margin (12.0%) and the largest Enterprise loss (-$175K). Government is the largest profit source in every country.

3.3 Discount bands

Discount band	Revenue	Profit	Margin	Discount rate
None	$7.94M	$1.74M	21.9%	0.0%
Low	$34.63M	$6.19M	17.9%	2.5%
Medium	$38.78M	$5.58M	14.4%	7.2%
High	$37.37M	$3.39M	9.1%	12.5%



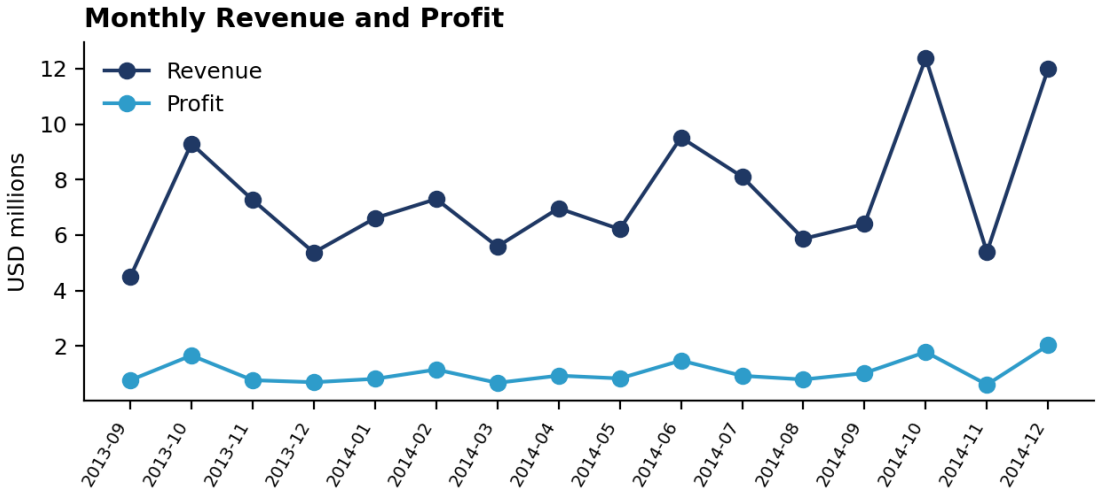
Margin falls steadily as discounts rise. In Enterprise, High-discount deals lose 10.0% of revenue (about -$0.62M), which explains most of the segment loss.

3.4 Products

Product	Revenue	Profit	Margin
Paseo	$33.01M	$4.80M	14.5%
VTT	$20.51M	$3.03M	14.8%
Velo	$18.25M	$2.31M	12.6%
Amarilla	$17.75M	$2.81M	15.9%
Montana	$15.39M	$2.11M	13.7%
Carretera	$13.82M	$1.83M	13.2%

Paseo is the leading product (28% of revenue and profit). Product margins are close (13-16%), so segment and discount level matter more than product choice.

3.5 Monthly trend



Revenue peaks in October and December. The data holds only four months of 2013, so full-year comparison would be misleading. On a like-for-like basis (Sep-Dec), revenue grew 36.9% and profit 40.1% from 2013 to 2014.

4. Strategic Recommendations

#	Recommendation	Why	Expected impact
1	Reprice Enterprise and set a minimum deal margin	Segment loses $0.61M on $19.6M revenue	Break-even alone adds about $0.6M profit
2	Require approval for high discounts	High band margin is 9.1% vs 17.9% for Low	Higher margin on 31% of revenue
3	Protect Government contracts and manage concentration risk	67% of profit comes from one segment	Secures the profit base
4	Grow Channel Partners and Midmarket through partner programs	Margins of 73% and 28%	Faster profit growth per dollar of sales
5	Plan stock, staff and promotions for Oct and Dec	Two months drive the peak	Better service and fewer stock-outs
6	Review US discounting and Enterprise mix	Lowest margin (12.0%), largest Enterprise loss	Lifts overall margin

5. Limitations and Next Steps

- The data covers 16 months, and 2013 has only four of them, so seasonality and growth should be confirmed with a longer history.
- Fractional values in Units Sold (36 rows) should be checked at the source.
- Findings show associations in this data. Test pricing and discount changes on a small scale before a full roll-out.
- Next step: build the interactive dashboard (see the Excel workbook) with slicers for Year, Segment, Country and Discount Band.